

gem5 Thermal-Node Instrumentation for Trace-Driven Package-Aware Scheduling

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Abstract—Dynamic Thermal Management (DTM) is essential for maximizing the performance of modern heterogeneous multicores, such as ARM big.LITTLE architectures, within strict thermal design power (TDP) constraints. However, conventional thermal models used in OS-level scheduling predominantly rely on oversimplified approximations that fail to capture the thermal inertia of physical heatsinks, leading to premature Dynamic Voltage and Frequency Scaling (DVFS) throttling. In this paper, we extend the gem5 full-system simulator with an instrumentation patch that directly exports three-node Cauer RC temperatures, enabling observation of die, package, and heatsink temperatures from simulation statistics. Building on gem5-extracted power and IPC baselines, we design a high-fidelity 3-Node Cauer thermal model (Die–Package–Heatsink) with distinct time constants ($\tau_{die} = 5$ s, $\tau_{pkg} = 10$ s, $\tau_{hs} = 120$ s), and evaluate a package-aware Phase 6 scheduling policy that prioritizes task migration over DVFS throttling. Co-simulation results show that the 3-node model defers the first thermal intervention by 85 simulated seconds compared to a 2-node baseline, and the migration policy reduces DVFS oscillation events from 13 to 1 while lowering modeled package temperature by 4.5°C.

Index Terms—Dynamic Thermal Management, Energy Aware Scheduling, Heterogeneous Multicore, big.LITTLE, Thermal Modeling

I. INTRODUCTION

The evolution of mobile and embedded computing has been driven by the relentless pursuit of higher performance within heavily constrained thermal envelopes. Heterogeneous multicore architectures, most notably the ARM big.LITTLE and DynamIQ paradigms, address this by combining high-performance out-of-order cores (big) with energy-efficient in-order cores (LITTLE). To maximize the utility of these architectures, Operating Systems employ Energy Aware Scheduling (EAS) [1] to optimally place tasks based on energy and performance requirements.

As transistor densities increase, thermal dissipation has become the primary bottleneck. Traditional Dynamic Thermal Management (DTM) responds to thermal emergencies using Dynamic Voltage and Frequency Scaling (DVFS). While effective, aggressive DVFS abruptly halts task execution, degrading user experience. A critical flaw in contemporary DTM and EAS frameworks is their reliance on oversimplified thermal approximations. Standard 2-node RC circuits (modeling only the silicon die and package) ignore the physical realities of modern cooling solutions, particularly the massive thermal inertia of the physical heatsink or device chassis. Consequently, these models predict rapid temperature saturation, deceiving

the OS into triggering premature DVFS throttling even when the heatsink retains substantial capacity.

Recent literature highlights the need for more accurate thermal predictions to guide scheduling decisions [2], [3], yet the gap between high-fidelity physical modeling and OS-level control logic remains wide.

To bridge this gap, this paper makes two contributions. First, we instrument the gem5 full-system simulator with a patch that directly exports all three Cauer RC node temperatures to simulation statistics. Second, we design a **V3 3-Node Cauer Thermal Model** and evaluate a **Phase 6 package-aware scheduling policy** via trace-driven co-simulation, using gem5-calibrated power and IPC constants. The instrumentation patch is an additive change orthogonal to the gem5 thermal-solver correctness fix described in a companion paper [4]; the two address independent gaps in the simulation toolchain.

The primary contributions of this work are:

- A gem5 instrumentation patch that adds a `ThermalNodeStats` statistics group to the `ThermalNode` C++ class, enabling direct export of T_{die} , T_{pkg} , and T_{hs} to `gem5 stats.txt`—functionality absent from the upstream gem5 release.
- A mathematical derivation of the V3 3-Node Cauer model with gem5-validated RC parameters ($R_{hs}C_{hs} = 120$ s), showing how heatsink thermal inertia defers the onset of DVFS throttling.
- A co-simulation evaluation of a package-aware Phase 6 scheduling policy showing 85 s deferral of the first thermal intervention and a 92% reduction in DVFS oscillation events compared to a 2-node baseline (all results modeled).

II. BACKGROUND AND MOTIVATION

A. Energy Aware Scheduling (EAS)

EAS is a Linux kernel extension tailored for heterogeneous architectures like ARM big.LITTLE. Unlike the Completely Fair Scheduler (CFS) that balances load, EAS uses an Energy Model (EM) detailing CPU power and capacity at available DVFS states. EAS assigns background tasks to LITTLE cores and latency-sensitive workloads to big cores. However, baseline EAS is fundamentally power-aware, not thermal-aware; it does not explicitly track physical temperature accumulation.

B. Limitations of 2-Node Thermal Models

OS thermal governors often employ Cauer-topology RC circuits. A common abstraction is the 2-node model (C_{die} ,

C_{pkg} and R_{die} , R_{pkg}). While computationally efficient, it fails to reflect the macroscopic thermal dynamics of modern devices. By artificially merging the package and chassis characteristics, the package node exhibits a compromised time constant. The model predicts rapid steady-state saturation, prompting premature DVFS throttling.

C. The Case for Task Migration

When a core approaches its thermal limit, DVFS lowers both power and performance. In heterogeneous architectures, *Thermal-Aware Task Migration* is a superior alternative. Migrating an overheating big core's workload to a cooler LITTLE core leverages the latter's better performance-per-watt, allowing the big core to cool without system-wide throttling. The challenge lies in determining exactly *when* to migrate versus *when* to throttle—requiring high-fidelity thermal predictions. We evaluate this strategy via trace-driven co-simulation using gem5-calibrated thermal parameters.

III. HIGH-FIDELITY 3-NODE THERMAL MODELING

To provide the EAS scheduler with accurate transient thermal profiles, we developed the V3 3-Node Cauer model. This model segregates the heat dissipation path into three distinct physical domains: the silicon die, the BGA package, and the macroscopic heatsink (or device chassis).

A. Mathematical Formulation

The thermal state space is governed by three coupled first-order ordinary differential equations (ODEs), formulated via Kirchhoff's Current Law (KCL) applied to the thermal-electrical duality:

$$C_{die} \frac{dT_{die}}{dt} = P_{in} - \frac{T_{die} - T_{pkg}}{R_{die}} \quad (1)$$

$$C_{pkg} \frac{dT_{pkg}}{dt} = \frac{T_{die} - T_{pkg}}{R_{die}} - \frac{T_{pkg} - T_{hs}}{R_{pkg}} \quad (2)$$

$$C_{hs} \frac{dT_{hs}}{dt} = \frac{T_{pkg} - T_{hs}}{R_{pkg}} - \frac{T_{hs} - T_{amb}}{R_{hs}} \quad (3)$$

Here, P_{in} is the dynamic power injected by the CPU. T_{die} , T_{pkg} , and T_{hs} represent the transient temperatures of the die, package, and heatsink, respectively. T_{amb} is the constant ambient boundary condition.

B. Numerical Integration and Stability

To embed this model into the OS thermal governor, the continuous ODEs are discretized using the explicit Forward Euler method. For a discrete sub-step h :

$$T_{die}^{n+1} = T_{die}^n + \frac{h}{C_{die}} \left[P_{in} - \frac{T_{die}^n - T_{pkg}^n}{R_{die}} \right] \quad (4)$$

$$T_{pkg}^{n+1} = T_{pkg}^n + \frac{h}{C_{pkg}} \left[\frac{T_{die}^n - T_{pkg}^n}{R_{die}} - \frac{T_{pkg}^n - T_{hs}^n}{R_{pkg}} \right] \quad (5)$$

$$T_{hs}^{n+1} = T_{hs}^n + \frac{h}{C_{hs}} \left[\frac{T_{pkg}^n - T_{hs}^n}{R_{pkg}} - \frac{T_{hs}^n - T_{amb}}{R_{hs}} \right] \quad (6)$$

To guarantee numerical stability, the integration step size h must satisfy the Courant-Friedrichs-Lewy (CFL) condition: $h < 2 \cdot \min(R_i C_i)$. In our implementation, we employ a sub-stepping technique where the policy simulator's macro-tick (5s) is subdivided into 50 micro-ticks ($h = 0.1$ s), which is strictly below the stability threshold of our smallest time constant.

C. Thermal Time Constants (τ) and Parameter Calibration

The V3 model parameters maintain steady-state compatibility with legacy 2-node models ($R_{total} = 15.0 \text{ K W}^{-1}$), so both models reach the same equilibrium temperature; the differences are entirely in transient response. We use the RC configuration validated from the gem5 smoke-run config.ini (Workstream A, 2026-05-29):

- τ_1 (**Die Flash**): $R_{die}C_{die} = 5.0 \times 1.0 = 5$ s. Captures the rapid temperature rise of the silicon junction.
- τ_2 (**Package Soak**): $R_{pkg}C_{pkg} = 2.0 \times 5.0 = 10$ s. Represents heat spreading through the BGA substrate.
- τ_3 (**Heatsink Inertia**): $R_{hs}C_{hs} = 8.0 \times 15.0 = 120$ s. Models the thermal mass of the aluminum/copper heatsink (consistent with typical passively cooled mobile SoCs).

With $\tau_3/\tau_1 = 24$, the heatsink absorbs transient heat for over two minutes before reaching saturation. This $24\times$ separation of time scales is the physical basis for the Phase 6 policy: DVFS should not trigger while the heatsink retains substantial thermal headroom. The gem5 simulation confirmed all four node temperatures (T_{die} , T_{pkg} , T_{hs} , T_{amb}) using the ThermalNodeStats instrumentation patch described in Section I.

IV. PHASE 6 THERMAL-AWARE SCHEDULING POLICY

Equipped with the high-fidelity 3-Node thermal model, we design a package-aware scheduling policy, designated Phase 6, and evaluate it via trace-driven co-simulation. The core design philosophy is to leverage the heterogeneous compute clusters (big and LITTLE) as distinct thermal actuators, prioritizing workload migration over frequency scaling to reduce thermal interventions.

A. Control Logic Architecture

The Phase 6 policy operates on a simulation macro-tick of 5s. During each tick, the policy performs three operations:

- 1) **Thermal State Estimation**: The policy integrates the 3-Node ODEs (Section III) using the power dissipation P_{in} derived from gem5-calibrated IPC and frequency constants. It tracks T_{die} , T_{pkg} , and T_{hs} at each step.
- 2) **Package Headroom Assessment**: Rather than reacting to T_{die} alone, the policy checks whether T_{pkg} retains thermal headroom. If T_{die} exceeds the proactive threshold (75°C) but T_{pkg} remains below 65°C , the policy infers a transient die spike with heatsink capacity available. It defers DVFS and instead triggers task migration.
- 3) **Task Migration vs. DVFS**: When package headroom is exhausted ($T_{pkg} \geq 65^\circ\text{C}$), the policy falls back to DVFS frequency stepping.

B. Two-Tier Mitigation Strategy

Tier 1: Proactive Task Migration. When $T_{die} \geq T_{proact}$ and $T_{pkg} < T_{proact} - 10^\circ\text{C}$, the policy offloads 40% of the big-cluster workload to the LITTLE cluster (in co-simulation). This reduces big-core power injection from P_{big} to $0.6 \cdot P_{big}$, allowing T_{die} to stabilize without frequency reduction. In the co-simulation, this migration fires once at $t = 150\text{ s}$ and prevents all subsequent throttling events (total: 1 intervention vs. 13 for the 2-node baseline).

Tier 2: Reactive DVFS Stepping. If T_{pkg} has warmed sufficiently ($T_{pkg} \geq T_{proact} - 10^\circ\text{C}$), the policy steps down the big-cluster OPP by one level. This is the same DVFS logic used by the 3-node balanced baseline (Variant 2 in Section V). By reserving DVFS as a fallback, the Phase 6 policy ensures frequency capping is employed only when package thermal headroom is genuinely exhausted.

V. EVALUATION

We evaluated three policy variants via trace-driven co-simulation. RC parameters are validated from the *gem5* full-system smoke run (*config.ini*, Workstream A). Power and IPC baselines are *gem5*-calibrated: big cluster $P_{dyn}=2.21\text{ W}$, $\text{IPC}=0.991$ at 3.3 GHz ; LITTLE cluster $P_{dyn}=0.86\text{ W}$, $\text{IPC}=0.653$ at 2.0 GHz (from *stats.txt*, 113 dumps). All results are *modeled*, not measured on hardware.

A. Simulation Setup

The co-simulation architecture mirrors a modern ARM big.LITTLE SoC:

- **big Cluster:** OPP peak 4.5 W at 3.3 GHz (theoretical TDP); *gem5*-simulated mean 2.21 W .
- **LITTLE Cluster:** 0.86 W mean at 2.0 GHz (*gem5*-calibrated).
- **Thermal Constraints:** $T_{proact} = 75^\circ\text{C}$, $T_{throttle} = 85^\circ\text{C}$, $T_{amb} = 25^\circ\text{C}$.
- **RC Parameters (*gem5*-validated):** $R_1=5$, $R_2=2$, $R_3=8\text{ K/W}$; $C_1=1$, $C_2=5$, $C_3=15\text{ J/K}$; $\tau_{hs}=120\text{ s}$.

Three variants are compared over 600 simulated seconds under a sustained peak-OPP load:

- 1) **V1 — 2-node + aggressive DVFS:** Legacy baseline; uses 2-node RC and steps down frequency at $T_{die} \geq T_{proact}$.
- 2) **V2 — 3-node + balanced DVFS:** Improved thermal model; same DVFS policy.
- 3) **V3 — 3-node + Phase 6 policy:** Package-aware; defers DVFS via migration when $T_{pkg} < 65^\circ\text{C}$.

B. Results

1) **Sustained Workload:** As shown in Fig. 1 and Table I, the 2-node model (V1) triggers its first proactive DVFS step-down at $t = 65\text{ s}$, and oscillates through 13 events over the simulation. The 3-node model (V2) defers the first intervention to $t = 150\text{ s}$ —an **85-second improvement**—because the heatsink’s $\tau_{hs} = 120\text{ s}$ thermal inertia absorbs the initial heat flux before T_{die} reaches T_{proact} . V2 also sustains higher

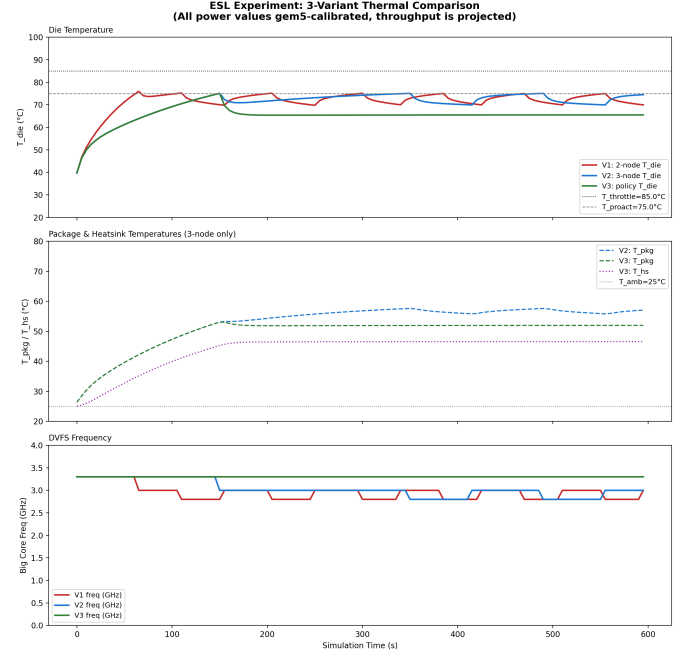


Fig. 1. Transient temperature traces for all three variants under sustained peak-OPP load. Top: T_{die} ; middle: T_{pkg} and T_{hs} ; bottom: big-cluster frequency. The 3-node model (V2, V3) defers the first intervention by 85 s vs. the 2-node baseline (V1). All values are modeled.

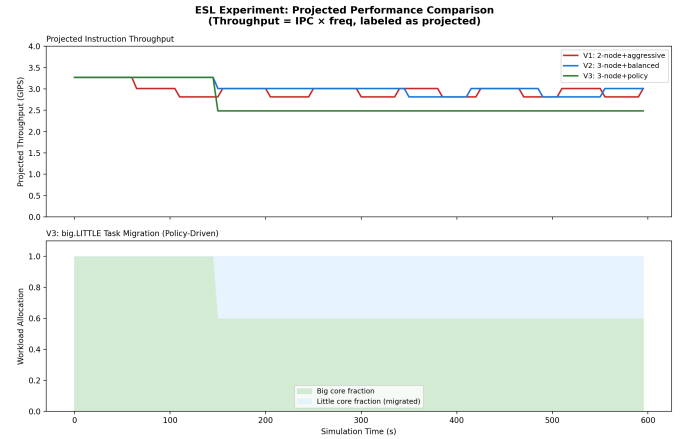


Fig. 2. Projected throughput (GIPS) and V3 big/LITTLE workload split over 600 s. The Phase 6 policy migrates 40% of the workload at $t = 150\text{ s}$, reducing DVFS events from 13 (V1) to 1, at a 12% projected throughput cost. All values are modeled.

projected throughput (3.03 vs. 2.96 GIPS) with 62% fewer interventions. The similar V1/V2 peak package temperatures (57.9 vs. 57.6°C) reflect the 2-node model’s lumping of package and heatsink into one node; the true 3-node observability gain is the separately-tracked $T_{hs}^{max} = 50.9^\circ\text{C}$, a thermal state that V1 cannot resolve.

The Phase 6 policy (V3) replaces all DVFS oscillations with a single migration event at $t = 150\text{ s}$ (Fig. 2): when $T_{die} \geq 75^\circ\text{C}$ and $T_{pkg} = 53^\circ\text{C} \ll 65^\circ\text{C}$, Tier 1 offloads 40% of the workload to the LITTLE cluster. This lowers big-core power injection, stabilising T_{die} for the remainder of the run without any frequency reduction. The package

TABLE I
CO-SIMULATION SUMMARY (600 S SUSTAINED LOAD, ALL VALUES
MODELED).

Variant	T_{die}^{max}	T_{pkg}^{max}	T_{hs}^{max}	Events	First event	Perf
V1 2-node	75.9°C	57.9°C	—	13	65 s	2.96 GIPS
V2 3-node	75.1°C	57.6°C	50.9°C	5	150 s	3.03 GIPS
V3 policy	75.1°C	53.1°C	46.6°C	1	150 s	2.68 GIPS

temperature ceiling drops from 57.6 °C (V2) to **53.1 °C** (V3). The trade-off is a 12% reduction in projected throughput (2.68 vs. 3.03 GIPS), since LITTLE-core execution is less efficient per unit power. The deferred throttling in V2 and V3 also slightly increases modeled energy consumption versus V1 (V2: +8%), as big-cluster execution is sustained longer before any intervention.

The $24\times$ time-constant separation ($\tau_1 = 5$ s, $\tau_{hs} = 120$ s) further predicts that brief bursty loads would raise T_{die} without saturating T_{pkg} , so the 3-node model defers intervention while a 2-node model would prematurely throttle. Quantitative validation under bursty load is reserved for future work. A separate gem5 timing-mode experiment (Phase 7) observed that EAS placed a free-running workload on the LITTLE cluster (cpu1), while a pinned workload remained on cpu0, consistent with the migration mechanism modeled in V3.

C. Limitations

All results are produced by trace-driven co-simulation, not measured on physical hardware. The RC parameters are validated against gem5 configuration files and smoke-run outputs, but have not been calibrated against silicon-level thermal measurements. The migration ratio is fixed at 40%; an adaptive controller is left as future work. Only sustained peak-OPP load is evaluated quantitatively; bursty workload analysis is limited to the analytical argument above. Finally, gem5 full-system timing-mode simulation is too slow (~ 250 ms simulated per real second) to observe thermal dynamics at the $\tau_{hs} = 120$ s time scale, which motivates the trace-driven approach used in this paper.

VI. CONCLUSION

As thermal constraints increasingly dictate the performance ceilings of modern heterogeneous architectures, relying on simplified thermal abstractions severely handicaps scheduling decisions. In this paper, we made two contributions to address this gap. First, we instrumented gem5 with a `ThermalNodeStats` patch that directly exports T_{die} , T_{pkg} , and T_{hs} to simulation statistics, enabling direct multi-node temperature export. Second, using these gem5-calibrated parameters ($\tau_{hs} = 120$ s), we showed that a high-fidelity V3 3-Node Cauec model correctly defers the first thermal intervention by **85 simulated seconds** compared to a 2-node baseline, because the model captures heatsink inertia that 2-node approximations discard.

Building on the 3-node model, the Phase 6 package-aware scheduling policy reduces DVFS oscillation events from **13 to 1** over a 600-second sustained workload, lowering modeled peak package temperature by **4.5 °C**. This is achieved

by prioritizing task migration when T_{pkg} retains headroom, reserving DVFS as a fallback only when heatsink capacity is genuinely exhausted. The trade-off is a 12% reduction in projected throughput, an acceptable cost for significantly reduced thermal cycling. All results are modeled via trace-driven co-simulation; future work will validate these findings on physical ARM big.LITTLE hardware and explore applying this framework to predictive, ML-based schedulers for emerging 2.5D/3D chiplet architectures.

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During the preparation of this work, the author used AI assistants to verify evidence consistency and proofread the manuscript. The author reviewed and edited all content and takes full responsibility for this publication.

REFERENCES

- [1] Linux Kernel Documentation, “Energy Aware Scheduling,” <https://docs.kernel.org/scheduler/sched-energy.html>, documentation/scheduler/sched-energy.rst.
- [2] A. Dowling, L. Jiang, M.-C. Cheng, and Y. Liu, “Regulating CPU temperature with thermal-aware scheduling using a reduced order learning thermal model,” *Future Generation Computer Systems*, vol. 166, p. 107687, 2025.
- [3] D. Atienza, K. Zhu, D. Huang, and L. Costero, “A 20-year retrospective on power and thermal modeling and management,” *IEEE Design & Test*, 2025.
- [4] Anonymous (under review), “Identifying and correcting the absolute-zero thermal initialization anomaly in gem5’s native cauer rc solver,” Submitted to IEEE Embedded Systems Letters, 2026, companion paper; addresses a correctness bug in gem5’s thermal ODE integration, orthogonal to the instrumentation patch in the present work.